

COVID-19 Severity: Predictive Modeling and Risk-Score Analysis

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1. Introduction

The goal is to identify demographic and clinical factors that predict severe COVID-19 illness and to derive a patient-level risk score. From the 10,000-participant study, 1,000 were drawn at random (`set.seed(6107)`) and partitioned 80 / 20 into training ($n = 800$) and test ($n = 200$) sets. The outcome is binary (1 = severe, 0 = not severe). All 13 candidate predictors in the codebook were retained — seven continuous (age, height, weight, BMI, SBP, LDL, depression) and six categorical, including the multi-level factors race and smoking.

2. Exploratory analysis

Severe outcomes occur in 32% of the training sample, with no missingness and tight, well-behaved distributions on the continuous predictors. Summary statistics by outcome are in **Table 1** and the marginal patterns in **Figure 1**. Three observations drive the modelling choices that follow. **First**, severe patients are older (median age 61 vs 59) and have higher BMI and SBP. **Second**, the empirical severity rate is 62% in unvaccinated vs 10% in vaccinated patients — by far the largest single-variable gap (the next, hypertension, is 41% vs 23%). **Third**, height, weight and bmi are mechanically related ($\text{BMI} = \text{weight} / \text{height}^2$, with $r = 0.70$ between weight and BMI and $r = -0.50$ between height and BMI); SBP and hypertension correlate at $r = 0.80$. We kept all 13 codebook predictors and let regularised / tree-based models handle the redundancy.

3. Model training

Nine candidates were trained with `caret::train` on the 800-row training set using **shared 10-fold CV folds** (`trainControl(index = createFolds(...))`) so that every model is compared on identical resamples. Tuning and ranking metric: area under the ROC curve (`metric = "ROC"`). `caret` dummy-coded factors automatically; the two SVMs additionally centred and scaled the predictors. The seed (6107) was set before the split, before fold construction, and before every train call. **Every complexity hyperparameter** — `lambda`, `nprune`, `min.node.size`, `n.trees`, `interaction.depth`, `shrinkage`, C , σ — **was selected strictly inside its grid**, so the cross-model comparison is apple-to-apple. Three secondary picks land at a family’s natural endpoint (GLMNET $\alpha = 1$, MARS `degree=1`, RF `mtry=1`); these are not signs of an under-extended grid but themselves substantive findings — the data prefer the simplest / most-regularised variant within each family.

The candidate set spans interpretable and flexible families: **logistic regression** (`glm`); **penalised logistic regression** (`glmnet`, $\alpha \in [0, 1]$ over 11 values, λ over 30 log-spaced values e^{-8} – e^{-1}); **GAM** with thin-plate smooths on the continuous predictors; **MARS** (`earth`, `degree` 1–3, `nprune` 2–18); **LDA**; **random forest** (`ranger`, `mtry` $\in \{1, \dots, 8\}$, `min.node.size` $\in \{2, 4, \dots, 16\}$); **Adaboost** (`gbm`, `distribution = "adaboost"`, `n.trees` $\in \{1,000, 2,000, 3,000, 4,000, 5,000\}$, `interaction.depth` 1–4, `shrinkage` $\in \{0.0005, 0.001, 0.003, 0.01, 0.03\}$, `n.minobsinnode` = 10); and two SVMs — **linear** (`svmLinear`, C on a 30-point log grid from e^{-3} to e^8) and **radial-basis** (`svmRadialSigma`, C on e^{-5} – e^5 and σ on e^{-8} – e^{-2} , both with 10–15 grid points). CV tuning curves are in **Figure 2**; selected hyperparameters are in **Table 2**.

4. Results

4.1 Predictive performance — and the boosting / SVM question

CV ROC and held-out test-set AUC, accuracy, sensitivity, specificity, and Brier are in **Table 3** and **Figure 3**; CV ROC distributions are in **Figure 2 (panel e)**. **The answer to the researchers’ first question is no — boosting and SVM do not beat the simpler methods.** On 10-fold CV, MARS and GLMNET tie at the top (mean ROC 0.877); AdaBoost is essentially tied (0.876), and SVM linear (0.873), SVM radial, GLM, LDA, and GAM all sit within ≤ 0.01 of the top; **random forest is the only model meaningfully weaker on CV** (0.857). Bonferroni-corrected paired tests on the shared folds (`caret::diff`) corroborate this: every boosting/SVM-vs-simple-method pair returns adjusted $p = 1.00$, and the smallest p across all 36 pairs is 0.12 (RF vs AdaBoost). On the test set

every AUC falls in the narrow band [0.900, 0.913], a spread of only 0.012 — the 95% bootstrap CIs (Table 3) overlap completely and are each ~ 0.09 wide. **The test-AUC ranking is therefore essentially within sampling noise at $n_{\text{test}} = 200$:** RF rises from worst on CV to nominal first on test, while MARS slips from tied-first to last, with neither move exceeding $\sim 1\%$.

Why don't the flexible methods help? With the four dominant predictors entering the response in nearly additive, threshold-linear form (see §4.3), there is no nonlinear-interaction signal for AdaBoost or the radial SVM to exploit; the extra flexibility chases noise. RF is penalised more — `mtry=1` with `min.node.size=10` is the most regularised configuration on its CV curve, and even that loses ~ 0.02 to the simple methods. RF and boosting also show the lowest test sensitivities at the default 0.5 cutoff (Sens = 0.26 and 0.67) despite their nominally top-ranked AUCs — a probability-calibration issue, not a discrimination gap, since both AUCs sit in the same narrow band as the simpler methods'.

4.2 Predictive risk score

MARS and GLMNET tie at the top of CV ROC. The CV-selected GLMNET fit ($\alpha = 1$, $\lambda = 0.0125$) is also closed-form but keeps seven non-zero linear predictors that act over each variable's entire range; MARS instead reduces to **five terms in four predictors** with hinge breakpoints that read directly as clinical thresholds — the effect is exactly zero below the breakpoint and rises linearly above it — so we use MARS to define $\hat{p}(\text{Severe} \mid x)$:

$$\text{logit}(\hat{p}_{\text{Severe}}) = -0.96 + 0.171 h(\text{age} - 60) + 0.353 h(\text{bmi} - 28.7) + 0.089 h(\text{SBP} - 118) - 3.32 \cdot \mathbf{1}[\text{vaccinated}],$$

where $h(u) = \max(u, 0)$, so each continuous risk factor is flat below its breakpoint and rises linearly above it. This is unusually transparent for a “machine-learning” model. The score spans [0.01, 0.99] on the test set. At the default 0.5 cutoff, test sensitivity is 0.76 and specificity 0.92 (test Brier 0.111); clinicians can re-thresh along the ROC curve to trade the two off. Calibration on the test set (**Figure 4**) is tight in the bottom five and top two deciles; deciles 6–7 (predicted 0.25–0.40) over-predict observed severity by 0.10–0.15 and decile 9 under-predicts by 0.15. With only 20 patients per bin these deviations are comparable to per-bin Monte Carlo noise.

4.3 Key predictors and their effects

Three importance measures — MARS term selection, AdaBoost relative influence (**Figure 5a**), and random-forest permutation importance — agree on the top predictors and on their order: **vaccination status, SBP, age, and BMI**, which together account for about 98% of AdaBoost's relative

influence. Hypertension is a near-duplicate of SBP ($r = 0.80$) and falls out of MARS once SBP enters; race, gender, smoking, LDL, and depression each contribute below 1% to AdaBoost and are never selected by MARS.

The partial dependence plots (**Figure 5b**) show *how predicted risk varies with* each predictor under AdaBoost, used here as an independent check on the MARS hinges — boosting and MARS share no inductive bias, so agreement is informative. Effects are model-implied associations, not causal:

- **Vaccination:** by far the strongest single signal. The PDP gap between unvaccinated and vaccinated is 0.44 in $\hat{p}(\text{Severe})$ (0.56 vs 0.13), echoing the empirical-rate gap (62% vs 10%) and the -3.32 MARS vaccine coefficient.
- **Age:** flat below ~ 58 , rising sharply between 58 and 65 to a plateau near 0.36 (consistent with the MARS hinge at 60).
- **BMI:** flat below ~ 27 , then rising steeply between 27 and 33 (PDP climbing from ~ 0.29 to ~ 0.35), aligning with the MARS hinge at 28.7.
- **SBP:** rising monotonically from ~ 120 mmHg, with the steepest segment between 125 and 140 (PDP rising from ~ 0.25 to ~ 0.38). This captures the same signal as binary hypertension, which is why hypertension drops out of MARS once SBP is in the model.

These directions are biologically plausible and consistent across MARS hinge coefficients, AdaBoost PDPs, and the GAM smooths (not shown). They should be read as predictive associations rather than causal effects: vaccinated and unvaccinated patients differ systematically in age and comorbidities, so unadjusted comparisons overstate any causal effect of vaccination.

5. Conclusion

Q1. Boosting and SVM provide **no advantage** over simpler models on this sample: CV ROCs cluster in 0.87–0.88 with non-significant Bonferroni differences, and test AUCs all sit in [0.900, 0.913] with overlapping bootstrap CIs. **Q2.** We adopt **MARS** for parsimony (4 predictors vs GLMNET’s 7) and hinge breakpoints aligned to clinical thresholds; the five-term closed-form risk score (§4.2) is computable at the bedside. **Q3.** The dominant predictors — **vaccination status (strongly inversely associated), age, BMI, and systolic blood pressure** — enter additively in threshold-linear form, which is why simple methods suffice. **Limitations.** Single 1,000-patient observational sample, no external validation; coefficients are predictive associations, not causal effects; mid-risk and decile-9 calibration are uncertain at $n_{\text{bin}} = 20$.

Appendix — Tables and Figures

Table 1: Patient characteristics by COVID-19 severity (training set, n = 800). Continuous variables: mean (SD); categorical variables: n (column %).

	Severe (n = 253)	Not severe (n = 547)
Continuous, mean (SD)		
age	61.3 (4.4)	59.6 (4.4)
bmi	28.0 (2.7)	27.3 (2.6)
SBP	132.8 (7.5)	129.0 (7.9)
height	169.7 (6.4)	170.6 (5.9)
weight	80.5 (6.8)	79.3 (6.9)
LDL	111.9 (20.3)	109.3 (20.0)
depression	7.1 (1.9)	7.1 (2.0)
Categorical, n (column %)		
<i>vaccine</i>		
Yes	46 (18%)	420 (77%)
No	207 (82%)	127 (23%)
<i>gender</i>		
Male	126 (50%)	278 (51%)
Female	127 (50%)	269 (49%)
<i>race</i>		
White	153 (60%)	348 (64%)
Asian	19 (8%)	34 (6%)
Black	48 (19%)	109 (20%)
Hispanic	33 (13%)	56 (10%)
<i>smoking</i>		
Never	153 (60%)	344 (63%)
Former	75 (30%)	160 (29%)
Current	25 (10%)	43 (8%)
<i>hypertension</i>		
Yes	160 (63%)	233 (43%)
No	93 (37%)	314 (57%)
<i>diabetes</i>		
Yes	44 (17%)	75 (14%)
No	209 (83%)	472 (86%)

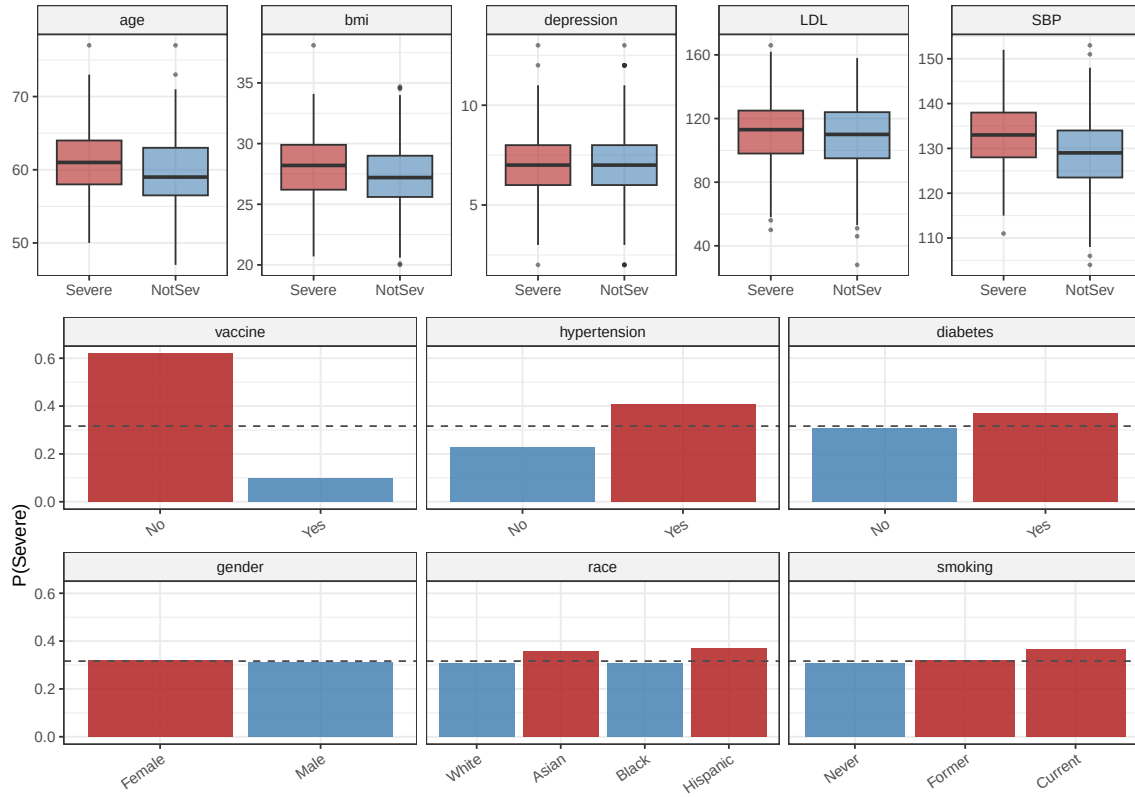


Figure 1: EDA — top: continuous predictors by severity (boxplots, training set, $n = 800$). Bottom: empirical severity rate by level for categorical predictors; bars above the overall rate of 32% (dashed line) are red, those below are blue.

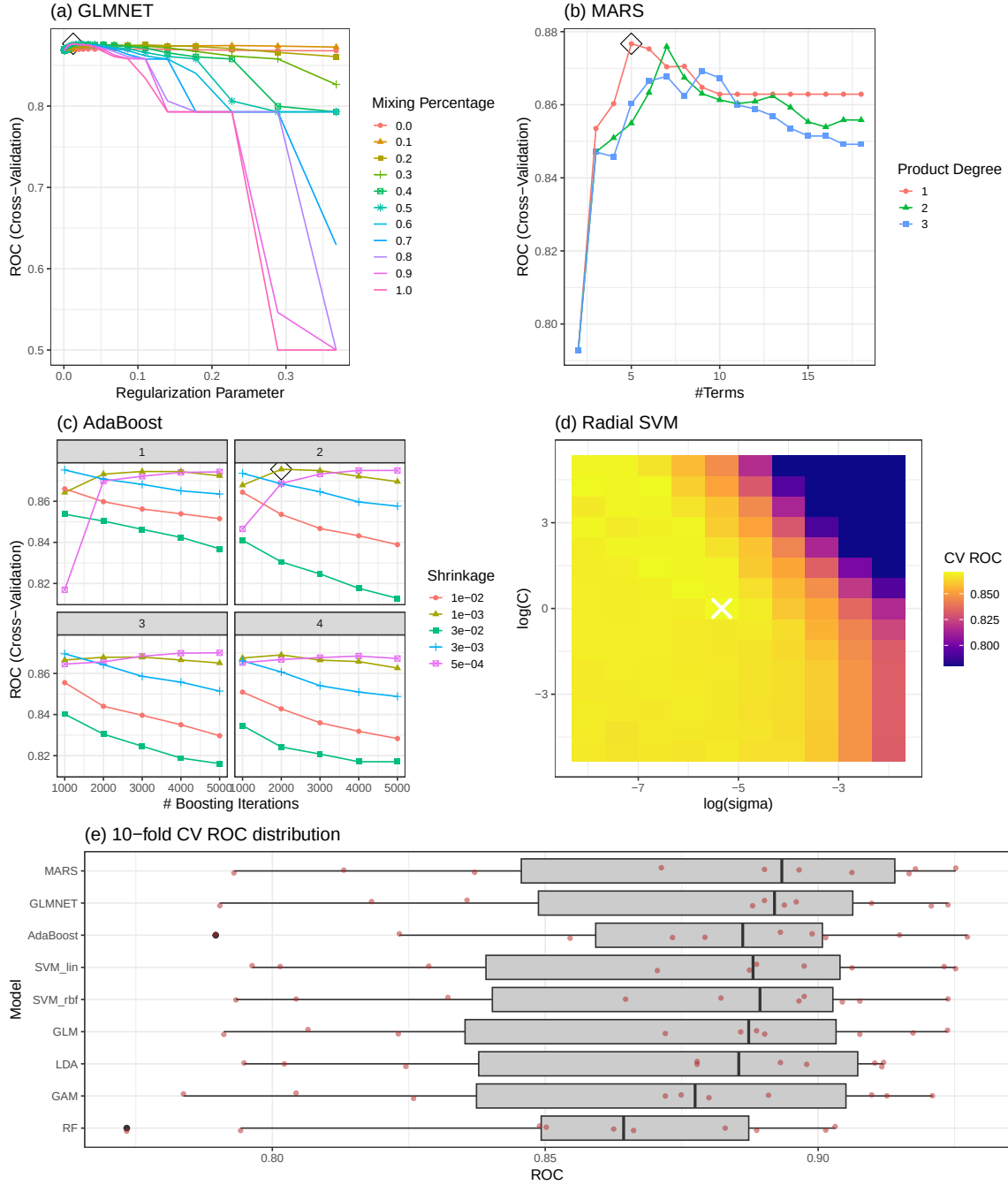


Figure 2: CV tuning curves and the resamples ROC distribution. (a) GLMNET ROC vs $\log \lambda$, coloured by mixing α . (b) MARS ROC vs $nprune$, coloured by interaction degree. (c) AdaBoost ROC vs $n.trees$, coloured by interaction depth, faceted by shrinkage. (d) Radial SVM ROC heatmap on $(\log \sigma, \log C)$; the cross marks the CV-selected pair. (e) 10-fold CV ROC distribution by model.

Table 2: Hyper-parameter tuning grids and CV-selected values. Continuous predictors are centred/scaled only for the two SVMs.

Model	Tuning grid (10-fold CV)	Best params
GLM	—	—
GLMNET	alpha: 11 pts in [0,1]; lambda: 30 log-pts in exp(-8..-1)	alpha=1, lambda=0.0125
GAM	default thin-plate smooths on continuous predictors	—
MARS	degree: 1–3; nprune: 2–18	nprune=5, degree=1
LDA	—	—
RF (ranger)	mtry: 1–8; min.node.size: {2,4,...,16}	mtry=1, splitrule=gini, min.node.size=10
AdaBoost (gbm)	n.trees: {1000,...,5000}; depth: 1–4; shrinkage: {5e-4, 1e-3, 3e-3, 0.01, 0.03}	n.trees=2000, interaction.depth=2, shrinkage=0.001, n.minobsinnode=10
SVM linear	C: 30 log-pts in exp(-3..8)	C=447
SVM radial	C: 15 log-pts in exp(-5..5); sigma: 10 log-pts in exp(-8..-2)	sigma=0.00483, C=1

Table 3: Cross-validated and test-set performance for all models, listed in the order they were introduced in lectures. CV ROC (10-fold mean) is the model-selection metric; test AUC is the held-out performance with 2,000-replicate stratified bootstrap 95% CIs (pROC: : ci . auc). Brier = mean squared error of predicted probabilities; lower is better.

Model	CV ROC (mean)	Test AUC (95% CI)	Test Acc.	Sens.	Spec.	Brier
GLM	0.871	0.909 (0.863, 0.950)	0.850	0.724	0.901	0.107
GLMNET	0.877	0.909 (0.859, 0.950)	0.860	0.759	0.901	0.107
GAM	0.868	0.910 (0.862, 0.948)	0.850	0.724	0.901	0.106
MARS	0.877	0.900 (0.850, 0.945)	0.870	0.759	0.915	0.111
LDA	0.870	0.909 (0.860, 0.949)	0.845	0.776	0.873	0.108
RF	0.857	0.913 (0.866, 0.952)	0.775	0.259	0.986	0.150
AdaBoost	0.876	0.906 (0.854, 0.948)	0.860	0.672	0.937	0.114
SVM_lin	0.873	0.907 (0.859, 0.949)	0.855	0.724	0.908	0.108
SVM_rbf	0.871	0.907 (0.856, 0.949)	0.835	0.741	0.873	0.109

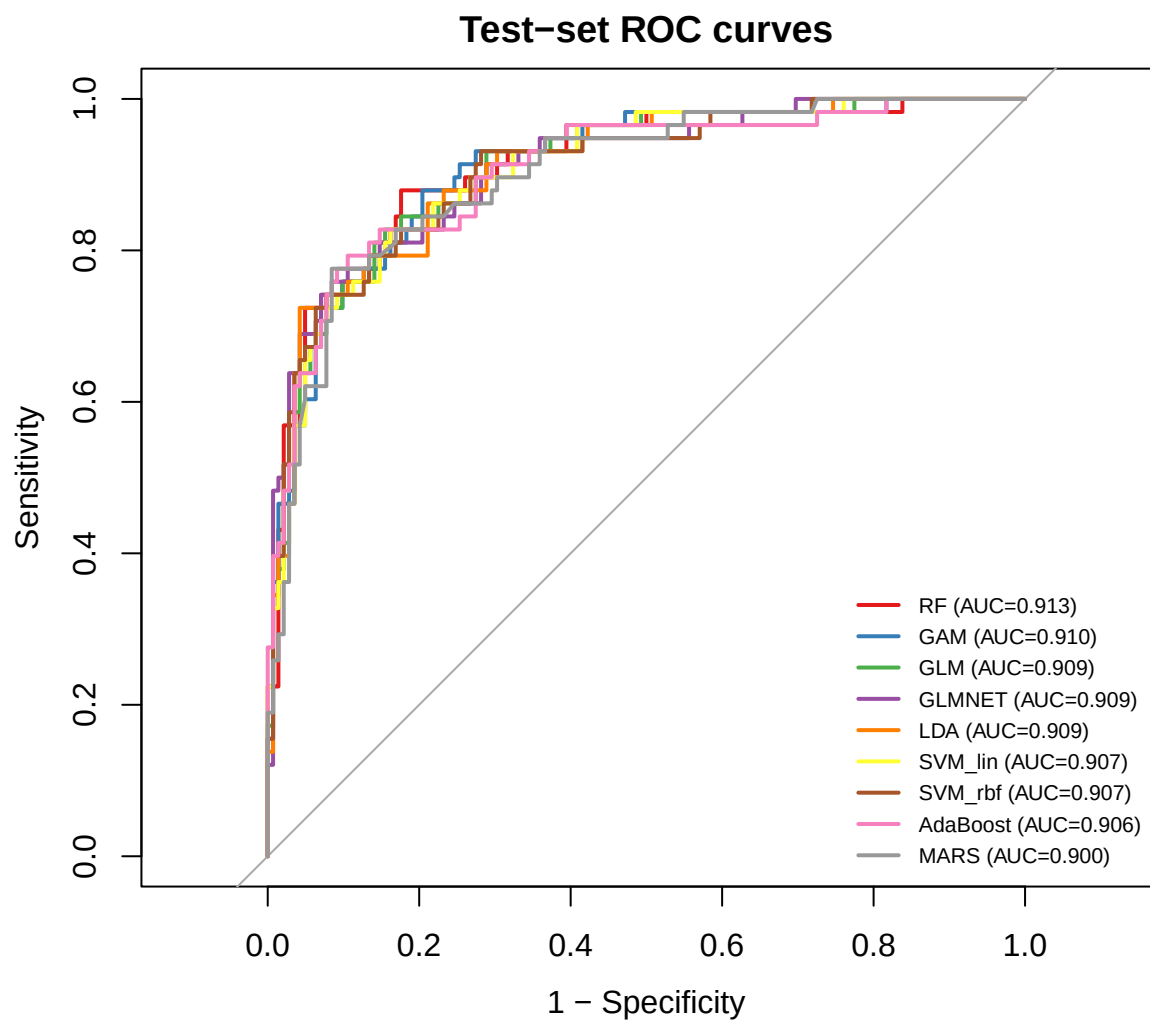


Figure 3: Test-set ROC curves for the nine fitted models with their AUCs.

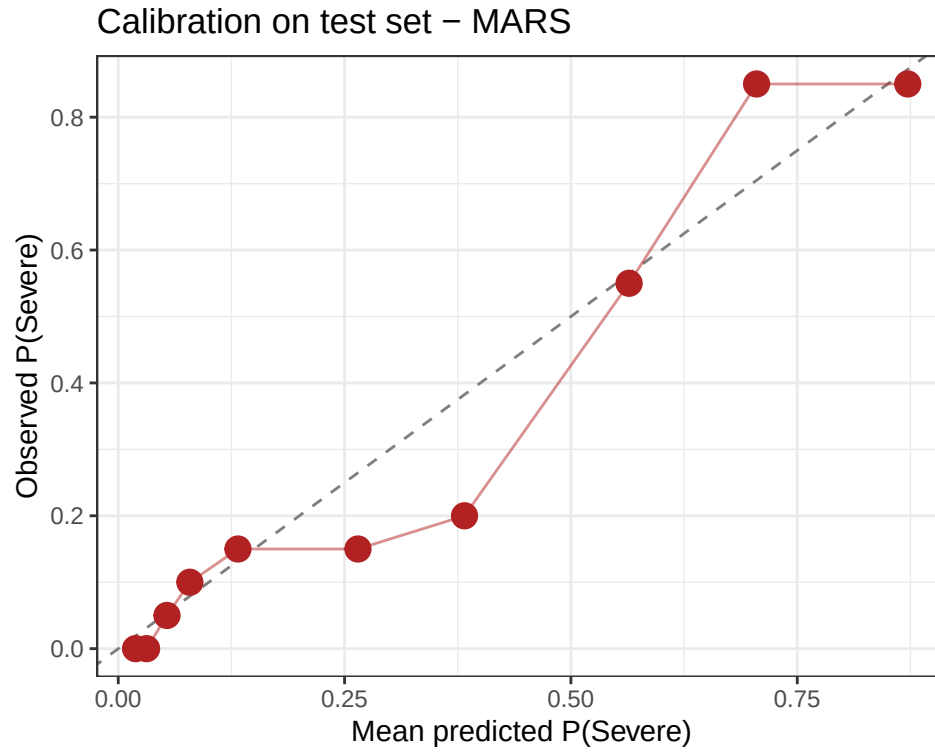


Figure 4: Calibration of the MARS risk score on the held-out test set. Patients are split into 10 equal-size bins by predicted P(Severe); each dot is a bin's mean predicted vs. observed severity rate (point area is bin size). Closer to the dashed diagonal is better.

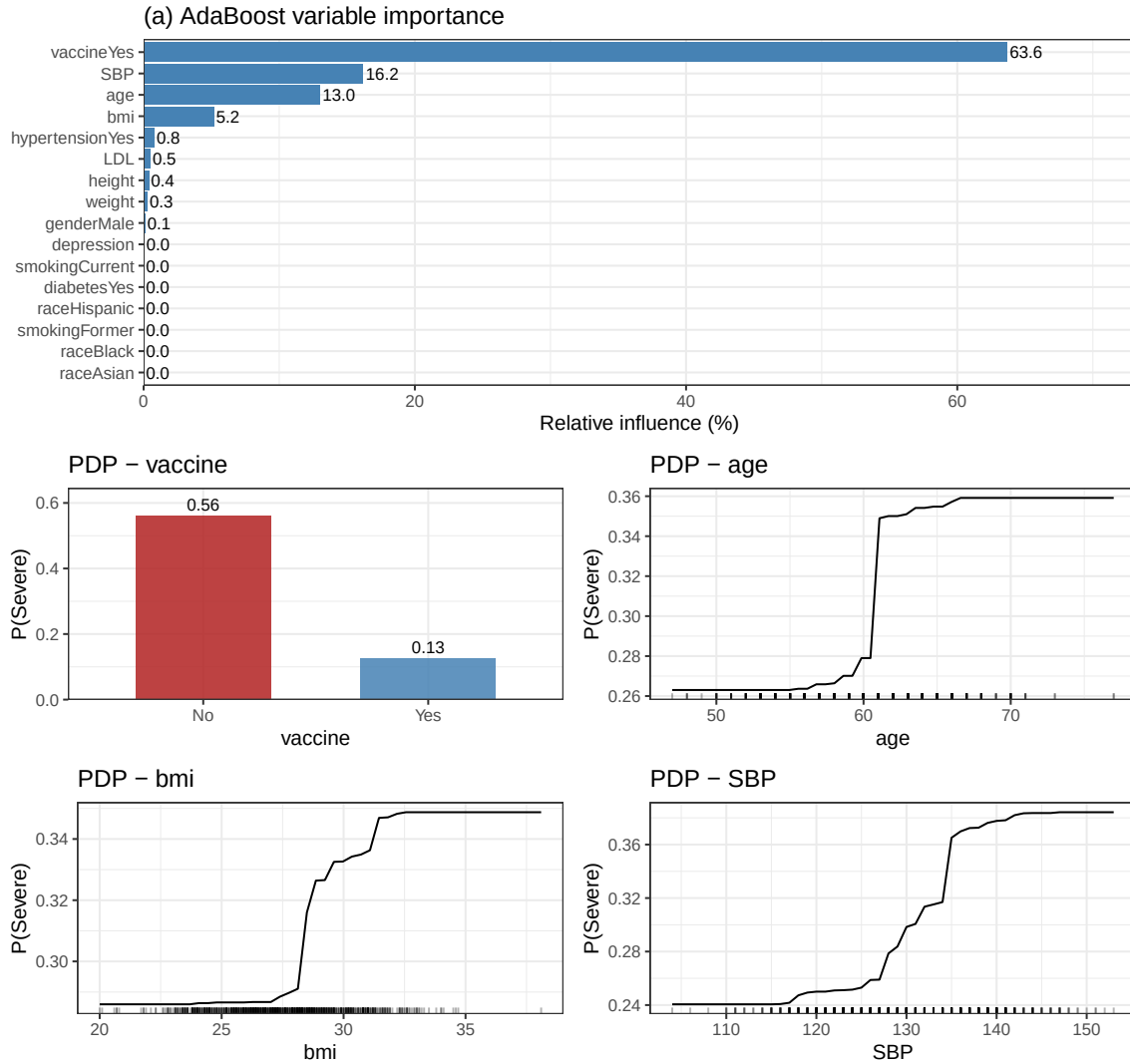


Figure 5: Boosting-based evidence on the four dominant predictors, used as an independent check on the MARS hinge structure. (a) Relative-influence variable importance from the AdaBoost final model — vaccine, SBP, age, and BMI together account for $\sim 98\%$ of the signal. (b) Partial dependence of $\hat{p}(\text{Severe})$ on each, holding the others at their training distribution; the elbow locations (~ 60 for age, ~ 28 for BMI, ~ 120 for SBP) match the MARS hinge breakpoints in §4.2.